Computer Science & IT



Compiler Design

Syntax Analysis (Parser)

Lecture No. 15



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Recap of Previous Lecture



Topic LALR(1) Parser

Topic LALR(1) Grammar

Topic Comparison between number of states

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Topics to be Covered



Topic Operator Precedence Parser



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Topic : Operator Precedence Parser

* It is a kind of bottom-up (shift-Reduce) parser.
* It is the only parser which can work with unambiguous as well as ambiguous grammar., but the grammar must be operator grammar
* Generally it is used to evaluate mathematical expression

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Topic: Operator Grammar

Operator grammar is a grammar with restrictions that

{ ① No two non-terminals should be adjacent
② No Epsilon production

① eg: $E \rightarrow E+E / E*E / T$
$T \rightarrow id$
Operator gramma (But ambiguous)

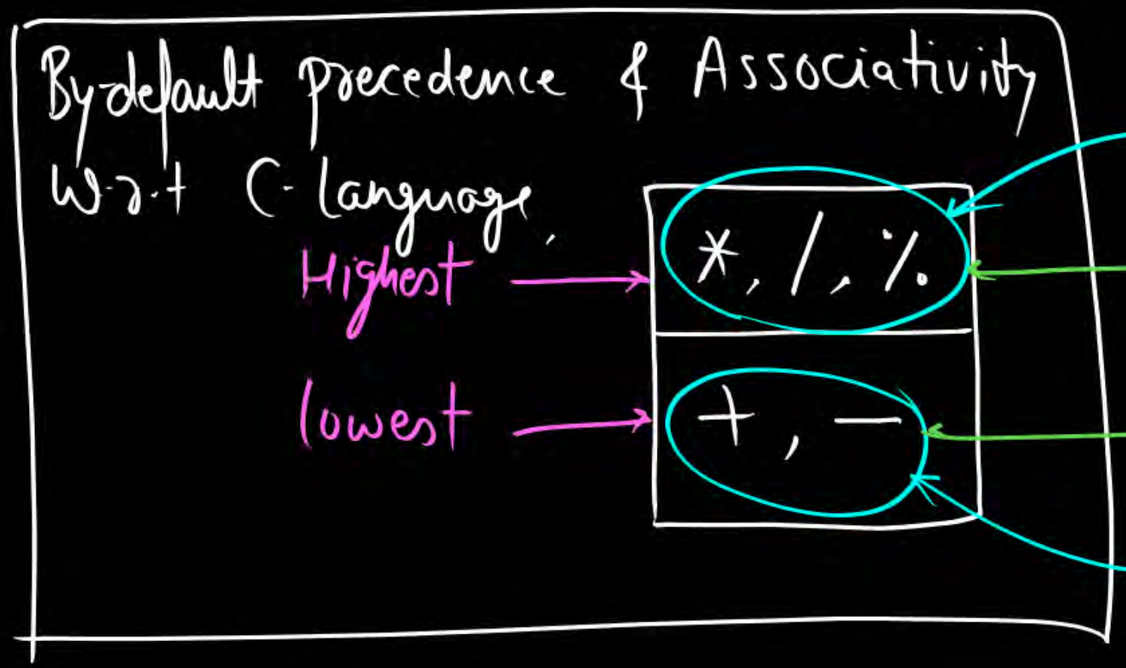
② eg: $E \rightarrow E+T / T$
$T \rightarrow T*F / F$
$F \rightarrow id$
Operator grammar Unambiguous

③ eg: $S \rightarrow ABC$
$B \rightarrow +/-$
$A \rightarrow a$
$B \rightarrow b$
Not an operator grammar
Adjacent Non-terminals

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* To evaluate any mathematical expression we need to know

{ ① Precedence of the operators { between different operators }
{ ② Associativity of the operator { Associativity is used for the operators of same precedence }



- If Operator grammar is Ambiguous, then we use general precedence and associativity defined by Programming language

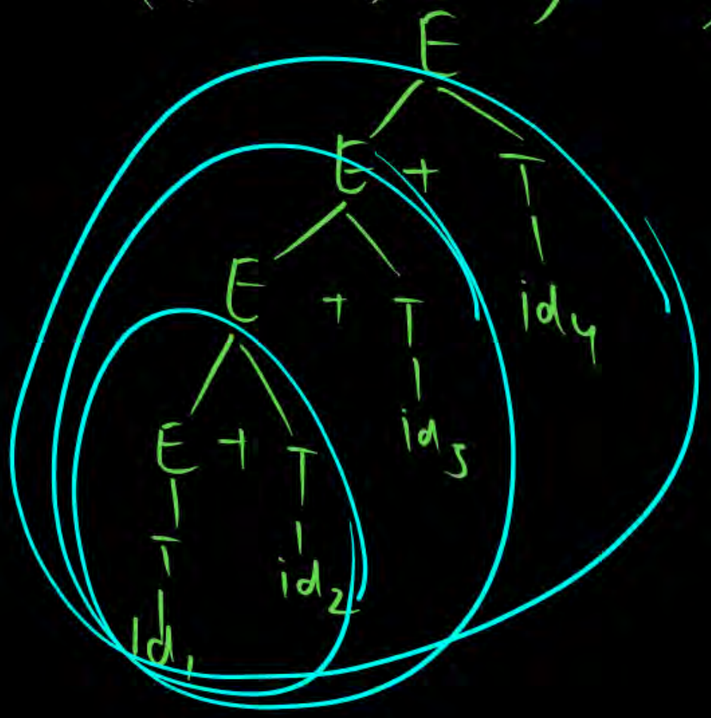
- If Operator grammar is Unambiguous, then grammar will define the precedence & associativity of the Operators.

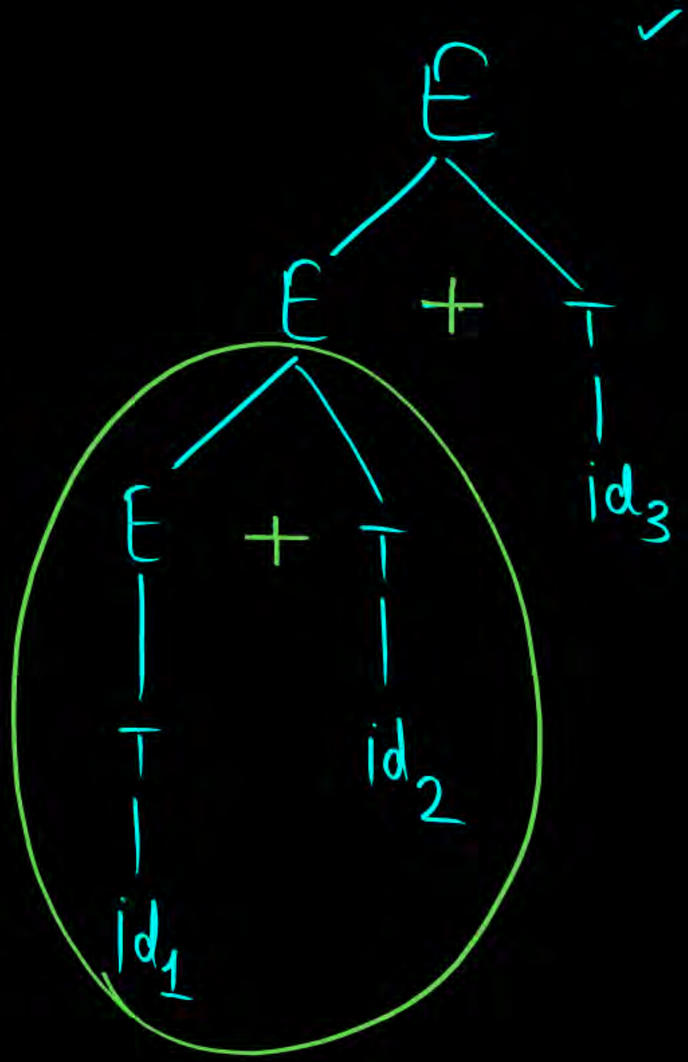
eg:-

$$E \longrightarrow E+T \mid T$$
$$T \longrightarrow id$$

$w = id_1 + id_2 + id_3 \$$

$(((id_1 + id_2) + id_3) + id_4)$

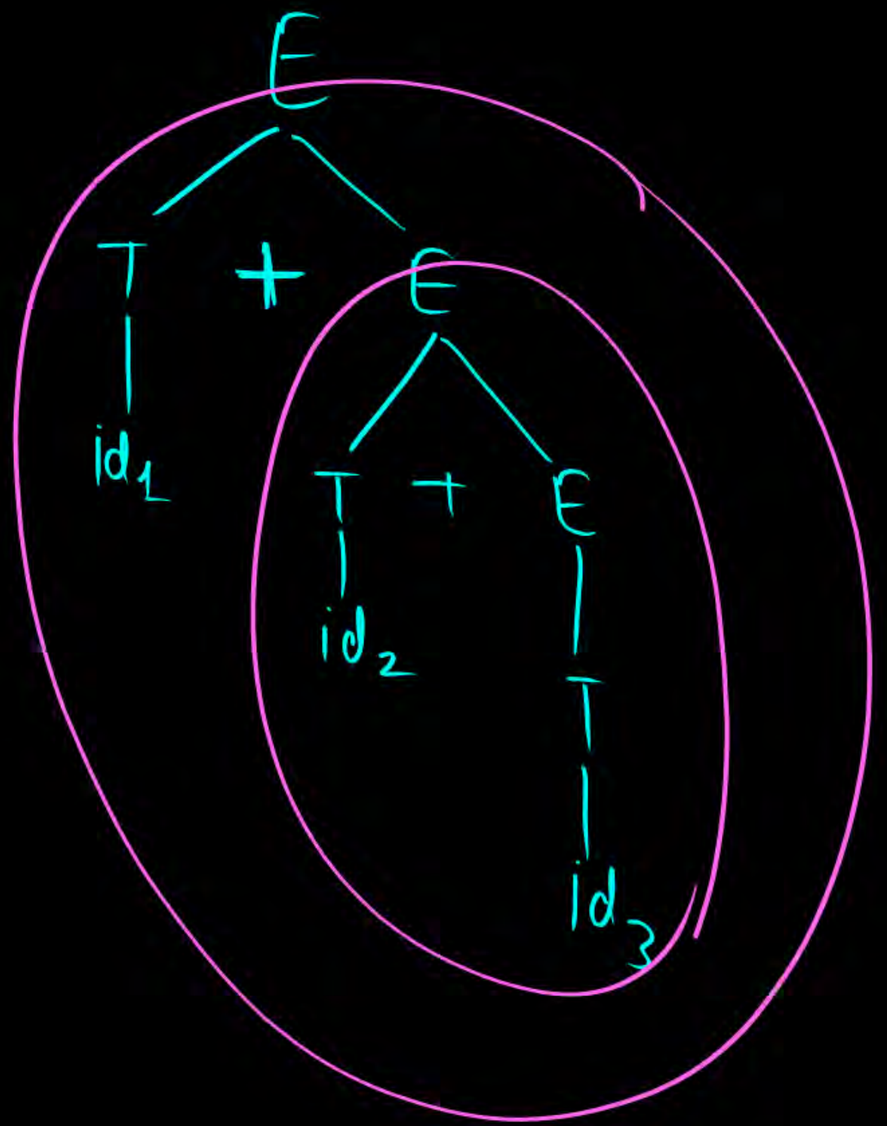




eg: $E \longrightarrow T + E / T$

$T \longrightarrow id$

$w = id_1 + (id_2 + id_3)$

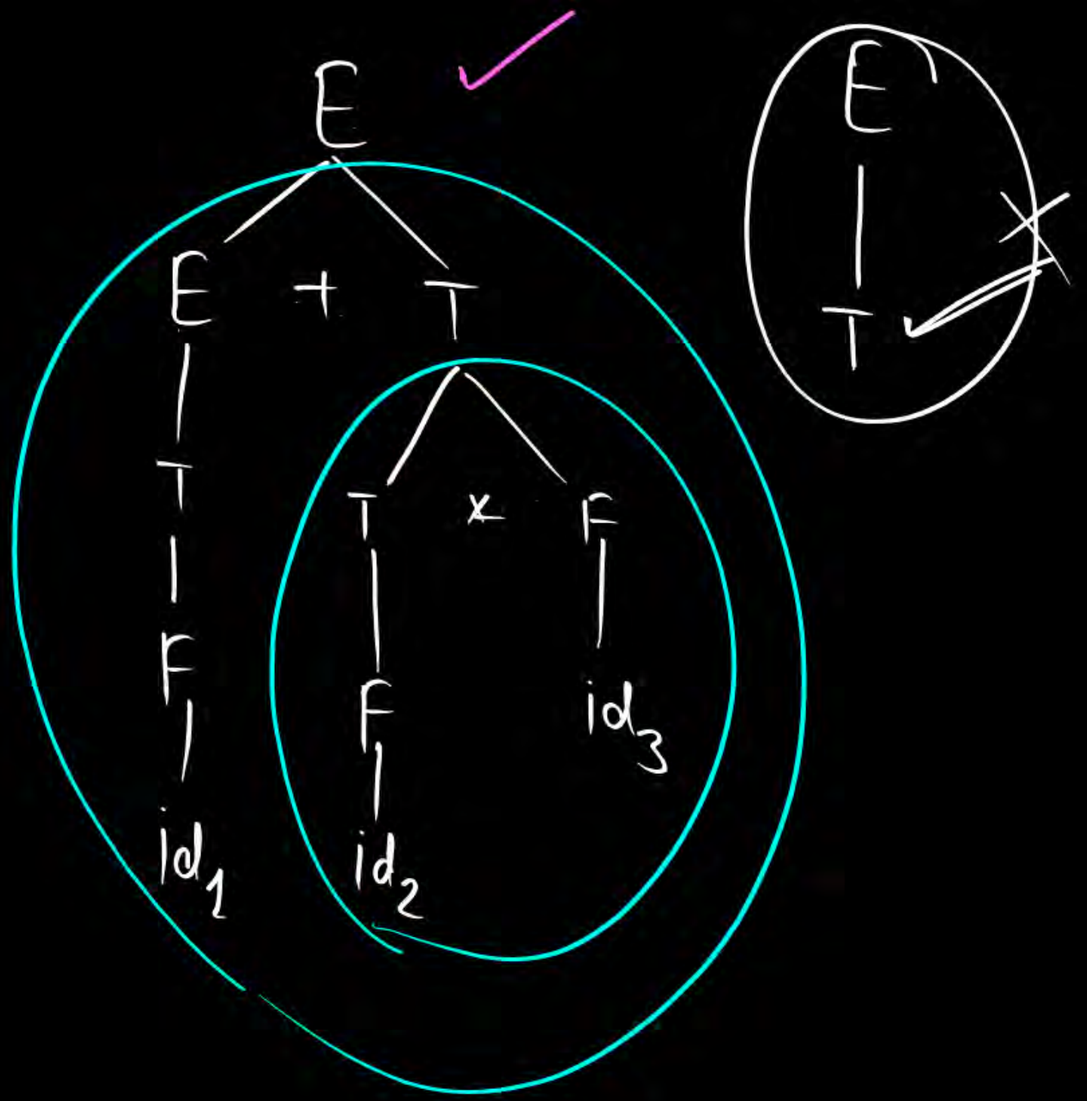


Note:- ① In an unambiguous grammar if Production corresponding to the operator is left-recursive, then that operator will be left-associative

② If production is right-recursive, then associated operation will be right-associative.

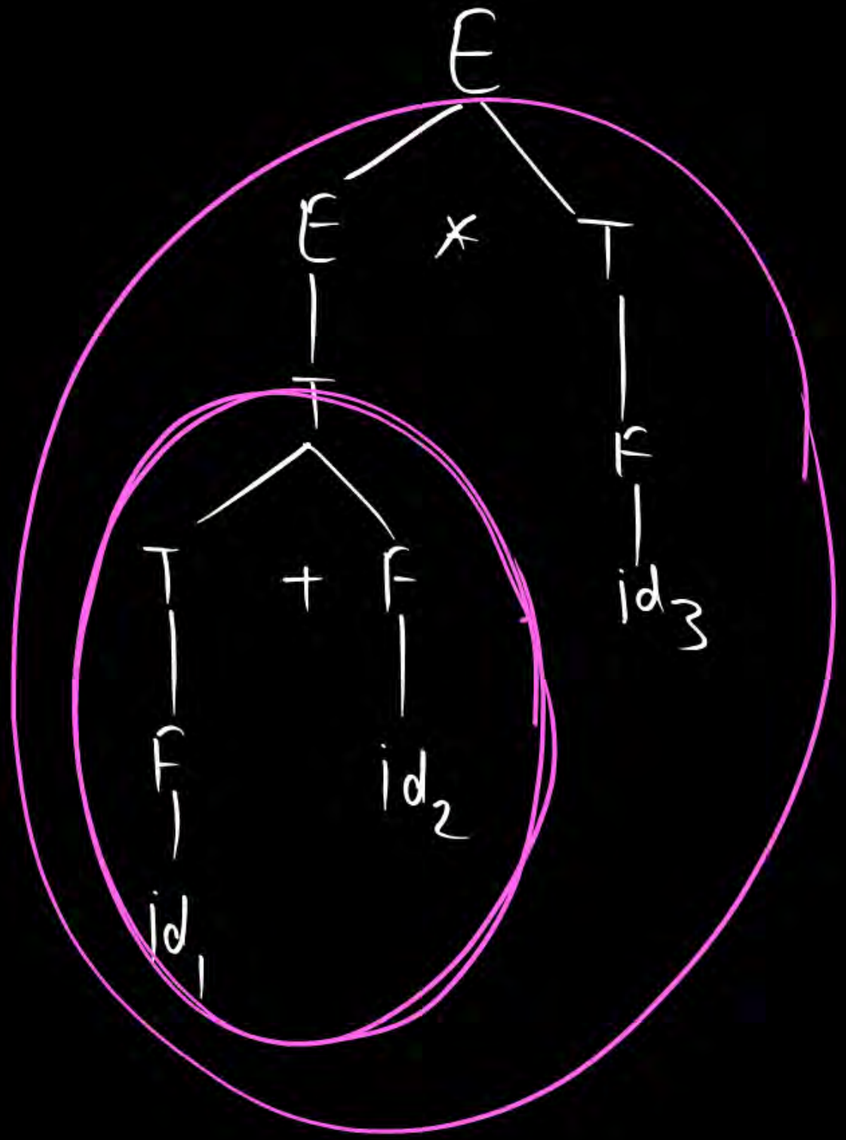
eg.
$$\begin{cases} E \longrightarrow E+T \,/\, T \\ T \longrightarrow T*F \,/\, F \\ F \longrightarrow id \end{cases}$$

$\omega = id_1 + id_2 * id_3 \$$



eg
$$\begin{cases} E \rightarrow E * T \mid T \\ T \rightarrow T + F \mid F \\ F \rightarrow id \end{cases}$$

$w = id_1 + id_2 * id_3$



Note:- In an unambiguous gramma precedence of the operator increases as we move down in the gramma.

Q: Find precedence order, and associativity of all the operators in the following grammar



① $E \rightarrow E * T / T$

$T \rightarrow F + T / F$

$F \rightarrow G - F / G$

$G \rightarrow G / H / H$

$H \rightarrow id$

What will be o/p w.r.t String $(5-(7-(9-2)))$ = 5

Precedence order :-

$/ > - > + > *$

Associativity

/ : Left Associative

− : Right Associative

+ : Right Associative

* : Left Associative

Q Consider the following grammar

$$\left\{\begin{array}{l} E \to E+E \mid T \\ T \to T*T \mid F \\ F \to id \end{array}\right\}$$

different Non-terminals

It is an ambiguous grammar,

- Precedece is clearly defined by the grammar,

  i.e., Precedence of * > Precedence of +

- Associativity is not defined by the grammar, {∴ we will use default associativity of + and *.}

Q: Consider the following gramman,

Same Non-terminals

$$\left\{\begin{array}{l} E \rightarrow E+E \\ E \rightarrow E*E \end{array}\right\} \equiv \left\{\begin{array}{l} E \rightarrow E+E \mid E*E \\ E \rightarrow id \end{array}\right\}$$

{Equal precedence based on given gramman}

$$\Downarrow$$

$$E \rightarrow E+E \mid E*E \mid id$$



Topic : Operator Precedence Table

Operation Relation table

→ Operator precedence parser uses the operator Precedence table for the purpose of parsing

$2 + 3 * 5 \$$

eg: $E \rightarrow E * T / T$ (left Associative)

$T \rightarrow F + T / F$ (Right Associative)

$F \rightarrow id$

Always True
* id will have the highest precedence
* $ will have lowest Precedence

	id	*	+	$
id	Error	$\gtrdot$	$\gtrdot$	$\gtrdot$
*	$\lessdot$	$\gtrdot$	$\lessdot$	$\gtrdot$
+	$\lessdot$	$\gtrdot$	$\lessdot$	$\gtrdot$
$	$\lessdot$	$\lessdot$	$\lessdot$	Accept

on left hand side

on Right hand side

Operator precedence table

Precedence of + > * w.r.t gramma

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Q: Construct the operator precedence table for the following gramman

$E \rightarrow E/T/T$

$T \rightarrow T*F/F$

$F \rightarrow G\dot{-}F/G$

$G \rightarrow H+G/H$

$H \rightarrow id$

LHS: top of Stack \ R.HS / Pointed by input pointer	id	/	*	−	+	$
id	Error	$\dot{>}$	$\dot{>}$	$\dot{>}$	$\dot{>}$	$\dot{>}$
/	$\dot{<}$	$\dot{>}$	$\dot{<}$	$\dot{<}$	$\dot{<}$	$\dot{>}$
*	$\dot{<}$	$\dot{>}$	$\dot{>}$	$\dot{<}$	$\dot{<}$	$\dot{>}$
−	$\dot{<}$	$\dot{>}$	$\dot{>}$	$\dot{<}$	$\dot{<}$	$\dot{>}$
+	$\dot{<}$	$\dot{>}$	$\dot{>}$	$\dot{>}$	$\dot{<}$	$\dot{>}$
$	$\dot{<}$	$\dot{<}$	$\dot{<}$	$\dot{<}$	$\dot{<}$	Accept

✱ Operator Precedence Parsing algorithm :-

Let 'X' is symbol on top of the stack and
'Y' is the symbol of input string pointed by input pointer.

Step-1 :- Shift $ into the stack

Step 2 :- if precedence of $X <\cdot Y$ or $X \doteq Y$, then shift 'Y' into the stack and increment input pointer

Step 3 :- if precedence of $X \cdot> Y$, then reduce
Pop elements from the stack until
Precedence of top of stack $<\cdot$ precedence of last element Popped from stack

Step 4 :- if $X = Y = \$$ then Accept

Note if [X, Y] in table = Blank, then error.



2 mins Summary

Topic Operator grammar

Topic Operator Precedence table

Topic Evaluation of expression

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THANK - YOU

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